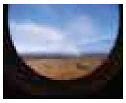
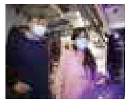


Mars on Earth
Cassandra Klos, a fine art photographer, simulates going to Mars with a team of experts. **https://dgit.in/dec21-41**



Five states of matter?
For the first time, an exotic condition of matter predicted about 50 years ago has been observed. **https://dgit.in/dec21-42**

Solar Kerosene

Working towards sustainable jet fuel

Paarth Pareek | feedback@digit.in

THE MINI SOLAR REFINERY

Made by a group of researchers led by Aldo Steinfeld who is the professor of renewable energy sources at ETH Zurich, the mini solar refinery has been operated by them on the roof of ETH's Machine Laboratory building in Zurich over the last two years. "This plant successfully demonstrates the technical feasibility of the entire thermochemical process for converting sunlight and ambient air into drop-in fuels. The system operates stably under real-world solar conditions and provides a unique platform for further research and development," says Mr. Steinfeld and he also thinks that the technology is now sufficiently mature for use in industrial applications.

IDEAL LOCATION

Now if we were to actually go ahead and set up plants to produce aviation fuels, the best location to do so would be in a desert region with high solar resources since it won't eat up agricultural land.

"Unlike with biofuels, whose potential is limited due to the scarcity of agricultural land, this technology enables us to meet global demand for jet fuel by using less than one percent of the world's arid land and would not compete with the production of food or livestock feed," explains Johan Lilliestam, a research group leader at the Institute for Advanced Sustainability Studies (IASS Potsdam) and professor of energy policy at the University of Potsdam.

HOW IT WORKS

According to Steinfeld, the hydrocarbon fuels developed in this process release only as much carbon dioxide during combustion as was previously extracted from the air during the manufacture. Here's how he says the process works at one of the said solar refineries in Móstoles, on the outskirts of Madrid.

An array of mirrors called a heliostat tracks the sun all the while boosting the sunlight's intensity by a factor of 2,500 while reflecting it on to a 50 foot high tower.

This concentrated beam of sunlight heats a reactor with a core made of cerium oxide which is an inexpensive compound often used to polish glass. At 2,700 degrees Fahrenheit, the oxygen gets liberated from the cerium and removed completely.

After which the water and carbon dioxide captured from the air are injected into the reactor and as it cools, the reduced cerium reaches

A solar refinery